TECHNICAL BULLETIN

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EPOXY CURING AGENT 100KA
(ECA 100KA)

Typical Properties

Property	Values	Test Method
Molecular Weight (average)	170-172	
Specific gravity @ 20° C, g/cc	1.166 ± 0.015	D-102 (ASTM D 4052)
Appearance	Clear to light yellow liquid	D-104 (Visual)
Viscosity at 25°C, cPs	100 - 200	D-150F
Gel Time, sec	140-190	D-160Y
Max Gel Temp, °F	375 Min	D-160Y
Color, Gardner	8 Max	D-150G

Applications

ECA 100KA is a low viscosity, light colored, non-crystallizing liquid anhydride epoxy curing agent based on methylhexahydrophthalic anhydride (MHHPA). For convenience, it is pre-catalyzed with an accelerator that contributes to good glass transition temperature (Tg), long pot life, and light color in cured polymers. This product is designed for use in converting epoxy resins to highly cross-linked polymers with excellent physical and electrical properties. ECA 100KA has the following characteristics:

- Non-crystallizing
- Good outdoor durability (UV resistance)
- Good alkaline resistance in pipe applications
- Low color of cured polymer

- Convenience of one component
- Low viscosity mixes with various epoxies for easy handling
- Long working time and pot life in epoxy systems

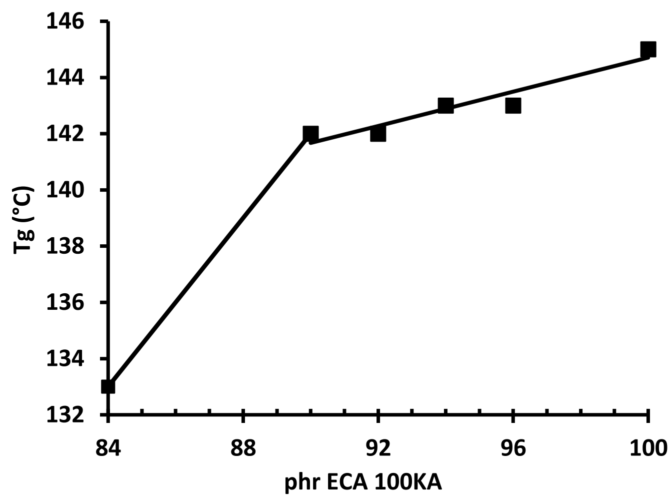
ECA 100KA imparts ease of handling due to its low viscosity, low volatility and low freezing point. If crystallization does occur during shipping or storage, it can be returned to liquid condition at 25°C or less. Formulations of ECA 100KA and liquid epoxies have a long pot life/working time in excess of 24 hours.

ECA 100KA is an excellent choice for curing liquid epoxy resins in demanding applications including laminating and filament winding.

For more information on the use of anhydrides like ECA 100KA as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical Company.

Anhydride Level

Liquid BPA epoxy resin was formulated with various levels of ECA 100KA. Samples were cured for 30 minutes at 100°C, the temperature was raised to 160°C over ten minutes, and the cure was continued for a total of 70 minutes. Glass transition temperatures (Tg) were measured, and are summarized in the following plot:



Epoxy Formulations

ECA 100KA was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes the results:

Epoxy type	ECA 100KA, phr	Viscosity at 25 °C, cP	Gel Time [1], min	Tg [2], °C
Standard BPA Liquid Epoxy	89	1560	33	144
Low Viscosity BPA Liquid	92	1430	32	142
Cycloaliphatic Epoxy	122	193	28	225
Epoxy Phenol Novolac	96	1530	33	142
Epoxy BPA Novolac	87	7270	32	145
BPF Liquid Epoxy	96	800	30	134

1. Shyodu Hot Pot (100g at 100°C).
2. Cured 1 hr at 120°C and 1 hr at 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

ECA 100KA will react with water to form diacids. This is normally undesirable, so ECA 100KA should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 100KA, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 100KA is available from Dixie Chemical Company, Inc.'s plant in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.